CTFS :

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

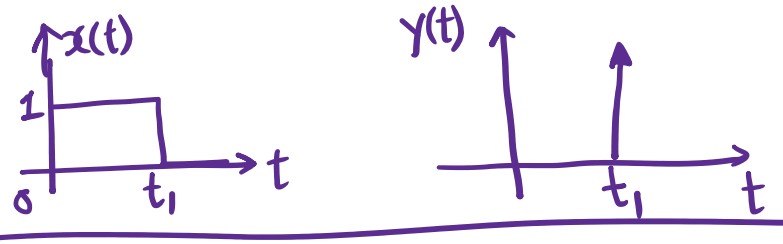
$$a_k = \frac{1}{T}\int_{\langle T\rangle} x(t) e^{-jk\omega_0 t}\, dt$$

Dirichlet Cond^n : (Sufficient but NOT Necessary)

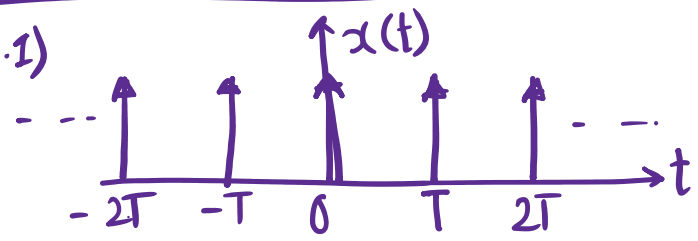
1) $\int_{\langle T\rangle} |x(t)| < \infty$

2) $x(t)$ must have finite no. of maxima & minima over a period.

3) " " finite discont. " "



Ex.1)

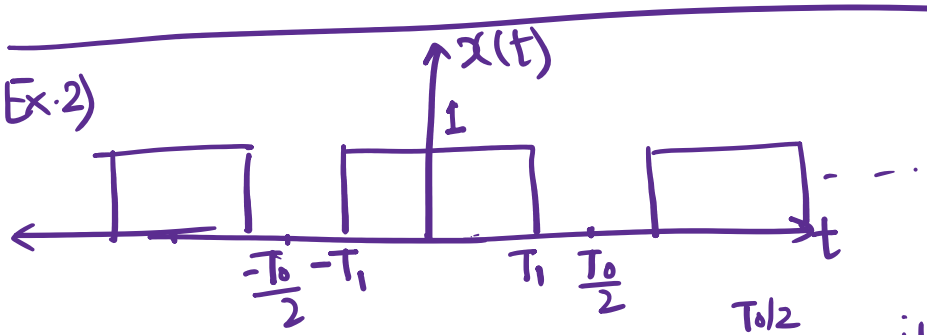


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad , \quad a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T}\int \delta(t).1\, dt = \frac{1}{T} \quad \forall k$$

$$\Rightarrow x(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

Ex.2)

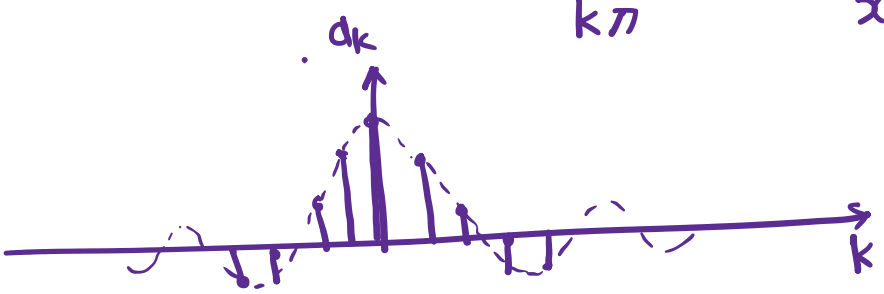


$$x(t) = \sum^{\infty} a_k e^{jk\omega_0 t} \quad a_k = \frac{1}{\ } \int^{T_0/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad a_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-T_1}^{T_1} 1 \cdot e^{-jk\omega_0 t}\, dt = \frac{e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}}{T_0 \cdot -jk\omega_0}$$

$$a_k = \frac{\sin k\omega_0 T_1}{k\pi} \qquad x_N(t) \triangleq \sum_{k=-N}^{N} a_k e^{jk\omega_0 t}$$



$$\sum_{k=-\infty}^{\infty} |a_k|^2 = \frac{1}{T}\int_{\langle T\rangle} |x(t)|^2 dt \longrightarrow \text{Parseval's Relat}^{n}$$

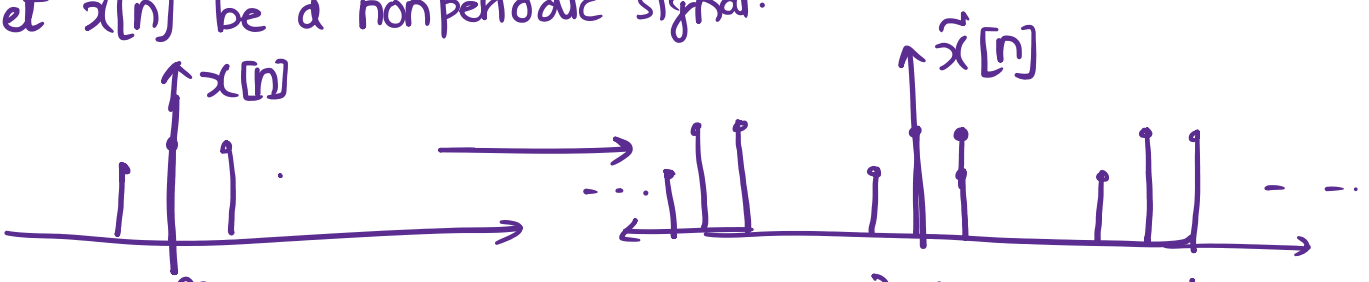
$$= \sum_k a_k \cdot a_k^* = \sum_k a_k \cdot \frac{1}{T}\int_{\langle T\rangle} x^*(t) e^{jk\omega_0 t}\, dt$$

$$= \frac{1}{T}\int x^*(t) \underbrace{\sum a_k e^{jk\omega_0 t}}_{x(t)}\, dt = \frac{1}{T}\int |x(t)|^2 dt$$

$a_{-k} = a_k^*$ (for real $x(t)$)

Discrete Time Fourier Transform

Let $x[n]$ be a nonperiodic signal.



$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n-kN], \quad N \text{ is suff. large} \quad \Big\} \ \tilde{x}[n] \text{ is periodic with period } N$$

$N-1$ $\quad jk\omega_0 n$ $\qquad$, $N-1$ $\quad -jk\omega_0 n$

$k=-\infty$ Period N

$$\tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} \quad , \quad a_k = \frac{1}{N}\sum_{n=0}^{N-1} \tilde{x}[n] e^{-jk\omega_0 n}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

$$a_k = \frac{1}{N}\sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}$$

Let $X(\Omega) \triangleq \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \Rightarrow a_k = \frac{1}{N} X(k\omega_0)$

$$\Rightarrow \tilde{x}[n] = \sum a_k e^{jk\omega_0 n} = \frac{1}{N}\sum_{k=0}^{N-1} X(k\omega_0) e^{jk\omega_0 n} \quad , \omega_0 = \frac{2\pi}{N}$$

$$= \frac{1}{2\pi}\sum_{k=0}^{N-1} X(k\omega_0) e^{jk\omega_0 n} \cdot \frac{2\pi}{N}$$

$$= \frac{1}{2\pi}\sum_{k=0}^{N-1} X\left(k\frac{2\pi}{N}\right) e^{jk\frac{2\pi}{N}n} \cdot \frac{2\pi}{N}$$

Now let $N \to \infty \Rightarrow \tilde{x}[n] = x[n]$, $\frac{2\pi}{N} \to 0$

$$= \frac{1}{2\pi}\int_0^{2\pi} X(\Omega) e^{j\Omega n} d\Omega = x[n]$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \quad \Big\} \text{ DTFT Analysis}$$

$$x[n] = \frac{1}{2\pi}\int^{2\pi} X(\Omega) e^{j\Omega n} d\Omega \quad \Big\} \text{ DTFT Synthesis}$$

Ex. 1) $x[n] = a^n u[n]$

$$X(\Omega) = \sum_{n=0}^{\infty} a^n e^{-j\Omega n} = \frac{1}{1-ae^{-j\Omega}} \quad , |a|<1$$

2) $x[n] = \delta[n]$, $X(\Omega) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\Omega n} = 1$